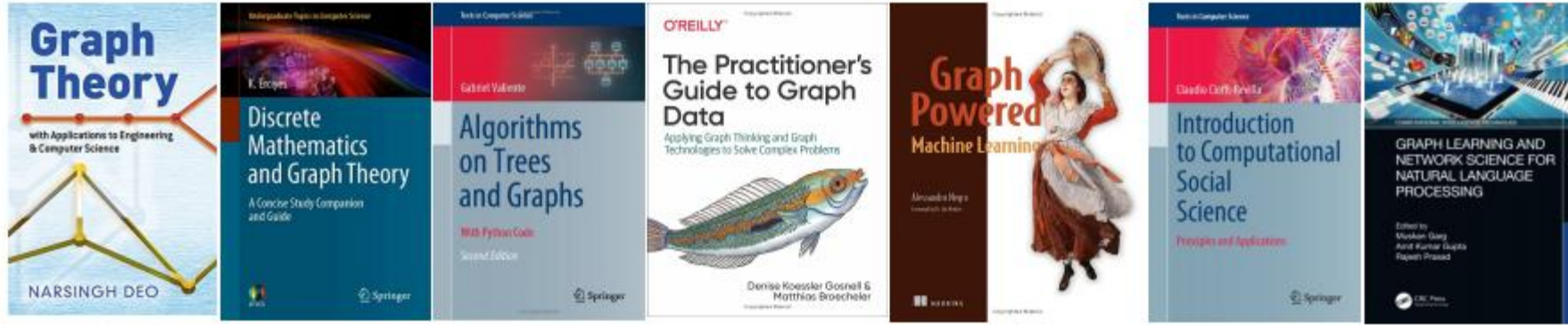


Chapter 10, Graphs

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- Spring 2026
- Aror University of Art, Architecture, Design & Heritage Sukkur



Graph Theory:

- Provide techniques for many clever data-structures for organizing information
- Many problems are modeled and efficiently solved using graphs
- particularly problems in Networks, Operating systems, Database

Graphs

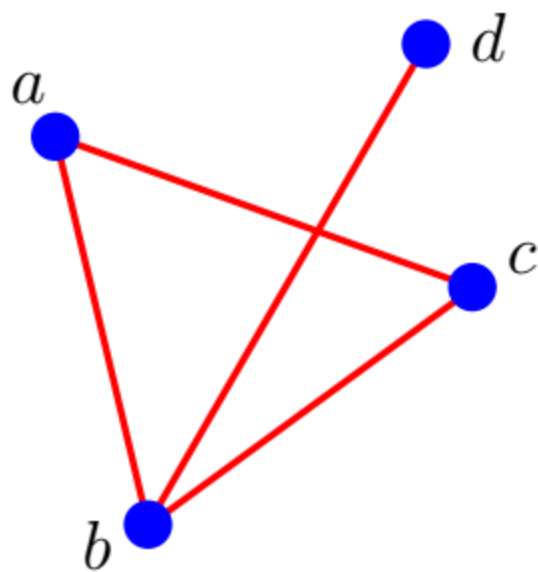
A graph is

- 1 a set of vertices V
- 2 a set of edges E ; each edge is a 2-subset of V

$$G = (V, E)$$

$$V = \{a, b, c, d\}$$

$$E = \{(a, b), (a, c), (b, c), (b, d)\}$$



Each edge is an unordered pair from V , we use the ordered pair notation

Types of Graphs: Undirected Simple Graphs

Applications require different types of graphs

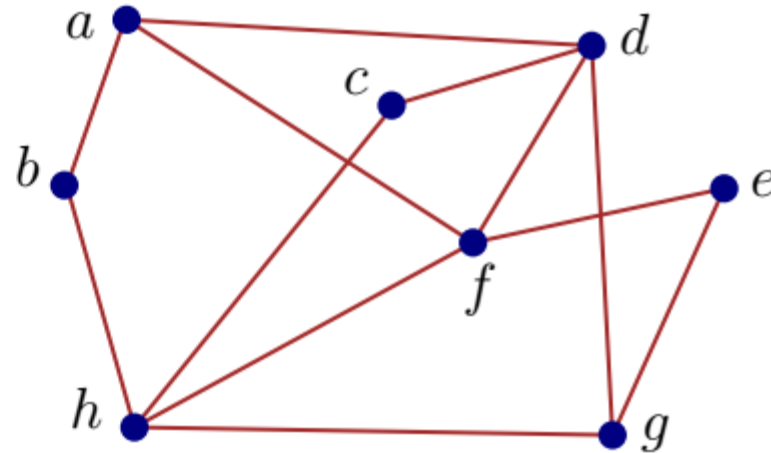
Undirected Simple Graphs

$$G = (V, E)$$

V is set of vertices

E is set of edges

(2-subsets of V , unordered pairs)



Types of Graphs: Undirected Simple Graphs

Let $G = (V, E)$ be a simple graph

Order of the graph is number of vertices in it, $|V| = n$

Size of the graph is number of edges in it, $|E|$

ICP 14-01

What is the maximum possible size of G of order n ?

$$\binom{n}{2}$$

Types of Graphs: PseudoGraphs and Multigraphs

Applications require different types of graphs

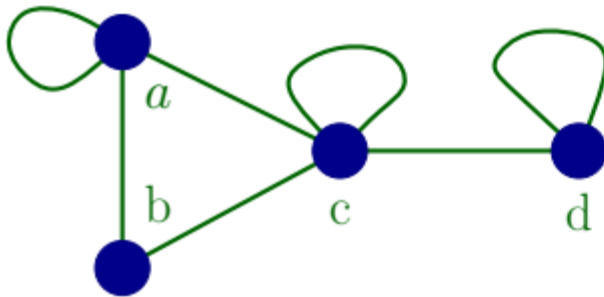
PseudoGraphs

$$G = (V, E)$$

V is set of vertices

E is set of edges

(self loops allowed)



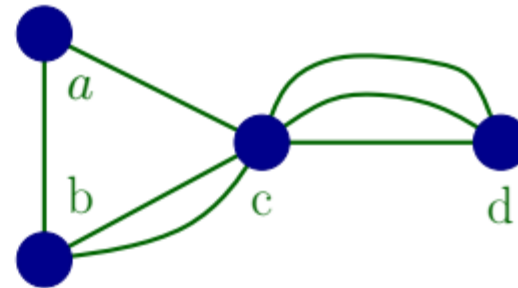
Multigraphs

$$G = (V, E)$$

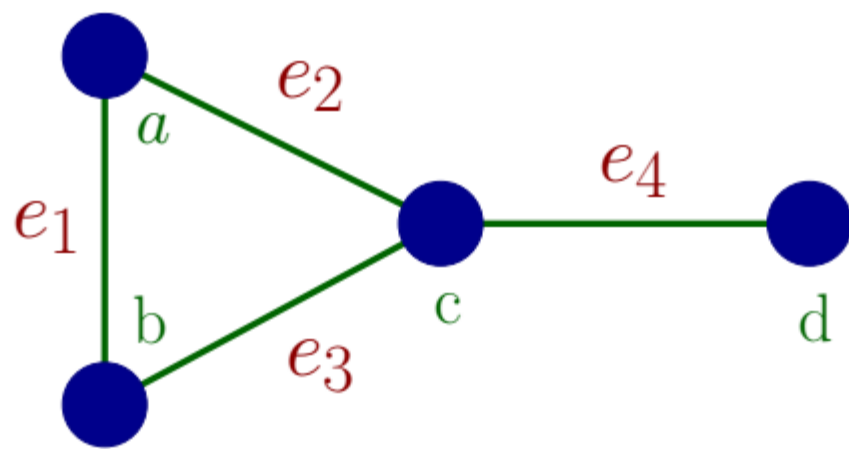
V is set of vertices

E is multi-set of edges

may have self loops too

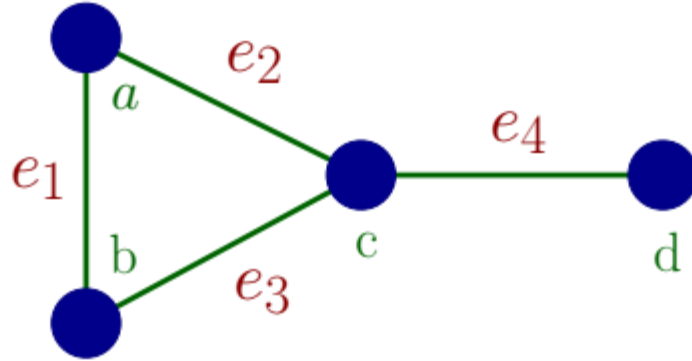


Graph Terminology: Incidence



- $e_1 = (a, b)$: a and b are endpoints of e_1 ▷ slight abuse of notation
- a and b are **adjacent**
- e_1 is incident to a and b

Graph Terminology: Degree



Degree of a vertex is the number of edges incident on it

The number of vertices adjacent to a vertex

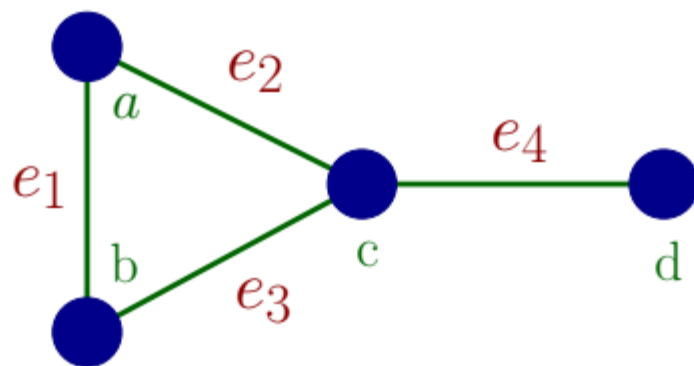
Denoted by $\text{deg}(v)$

ICP 14-02

What is the min and max possible degree of a vertex?

$$0 \leq \text{deg}(v) \leq n - 1$$

Graph Terminology: Degree



In this graph

ICP 14-03

$\deg(a) = ?$

ICP 14-04

$\deg(d) = ?$

ICP 14-05

$\deg(b) = ?$

ICP 14-06

$\deg(c) = ?$

Types of Graphs: Directed Graphs (digraphs)

Applications require different types of graphs

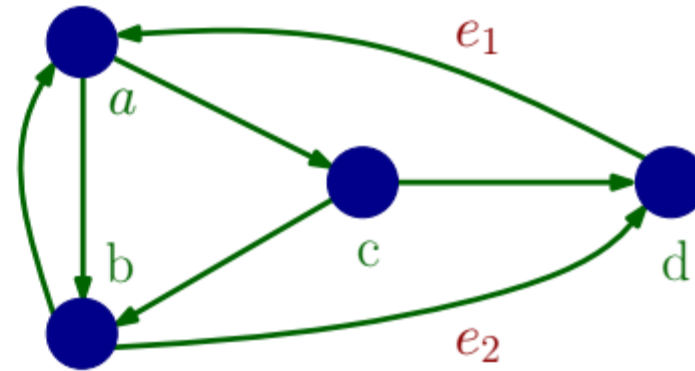
Directed Graphs (digraphs)

$$G = (V, E)$$

V is set of vertices

E is set of edges (ordered pairs)

$$E \subseteq V \times V \text{ (ordered pairs)}$$



Types of Graphs: Directed Graphs (digraphs)

Let $G = (V, E)$ be a simple digraph

Order of the graph is number of vertices in it, $|V| = n$

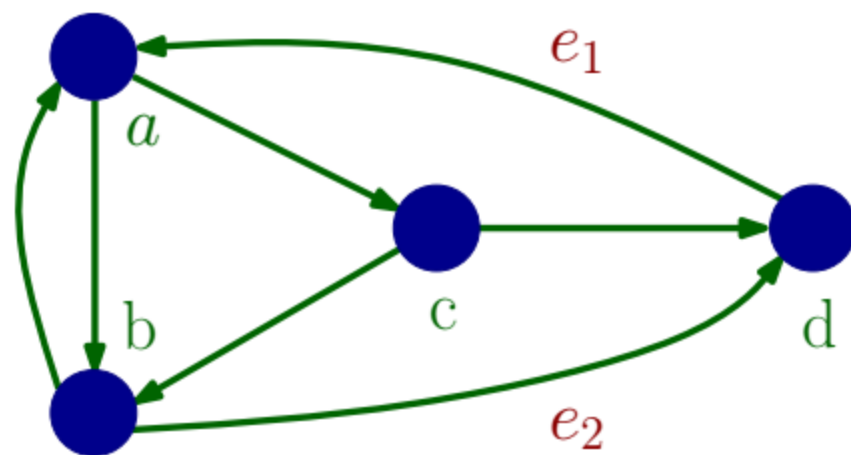
Size of the graph is number of edges in it, $|E|$

ICP 14-07

What is the maximum possible size of G of order n ?

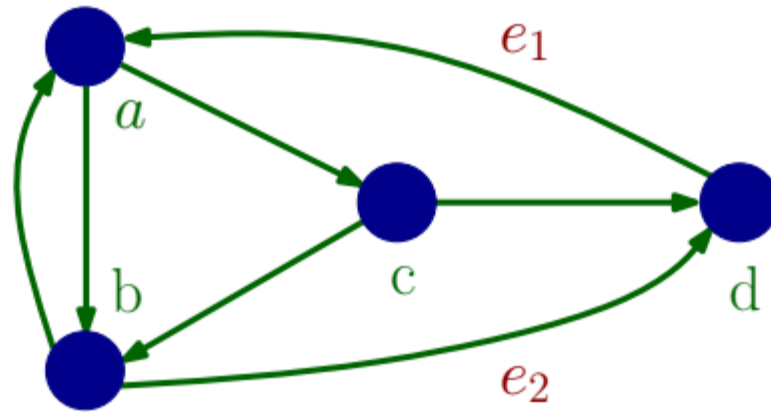
$$n(n - 1)$$

Graph Terminology: Directed Edges



- $e_1 = (d, a) \neq (a, d)$
- d is source of e_1 and a is target of e_1

Graph Terminology: In-degree and Out-degree



in-degree of a vertex is the number of (directed) edges incoming into it

The number of edges with this vertex as a target

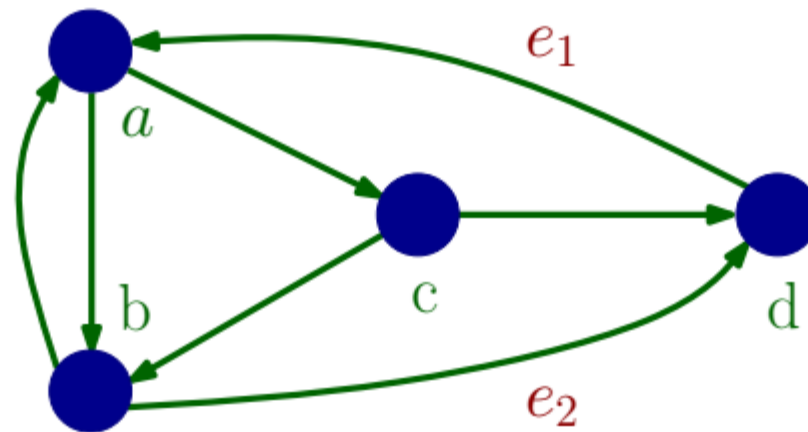
Denoted by $\deg^-(v)$ or $d^-(v)$

ICP 14-08

What is the min and max possible in-degree of a vertex?

$$0 \leq \deg^-(v) \leq n - 1$$

Graph Terminology: In-degree and Out-degree



out-degree of a vertex is the number of (directed) edges outgoing from it

The number of edges with this vertex as a source

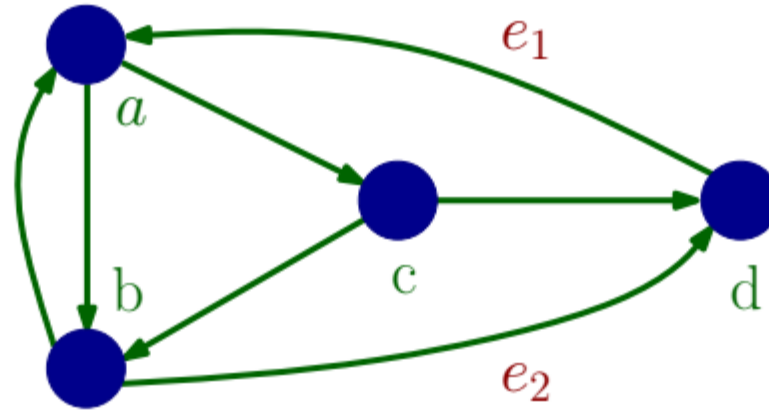
Denoted by $\deg^+(v)$ or $d^+(v)$

ICP 14-09

What is the min and max possible out-degree of a vertex?

$$0 \leq \deg^+(v) \leq n - 1$$

Graph Terminology: In-degree and Out-degree



In this graph

ICP 14-10

$$\deg^+(a) = ?$$

ICP 14-14

$$\deg^-(a) = ?$$

ICP 14-11

$$\deg^+(d) = ?$$

ICP 14-15

$$\deg^-(d) = ?$$

ICP 14-12

$$\deg^+(b) = ?$$

ICP 14-16

$$\deg^-(b) = ?$$

ICP 14-13

$$\deg^+(c) = ?$$

ICP 14-17

$$\deg^-(c) = ?$$

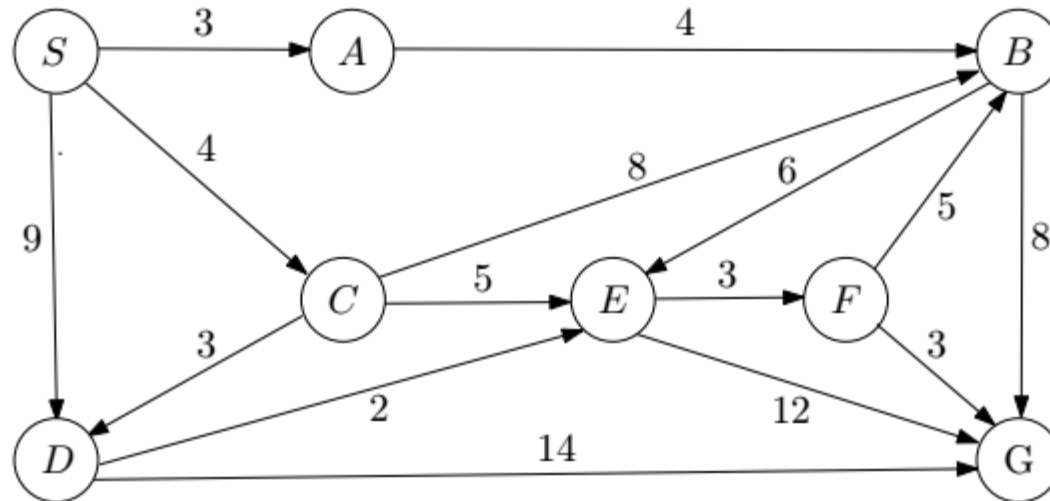
Types of Graphs: Weighted Graphs (digraphs)

Some applications work on graphs with weights on edges

Weighted Graphs (digraphs) : $G = (V, E, w)$

- V : Set of vertices
- E : Set of edges (directed edges)
- w : cost/weight function: $w : E \rightarrow \mathbb{R}$

▷ weights could be lengths, airfare, toll, energy



Graph Representation: Adjacency Matrix

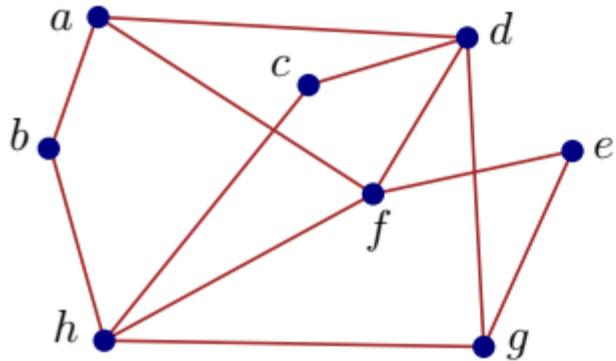
Undirected Simple Graphs

$$G = (V, E)$$

V is set of vertices

E is set of edges

(unordered pairs (2-subsets) of V)



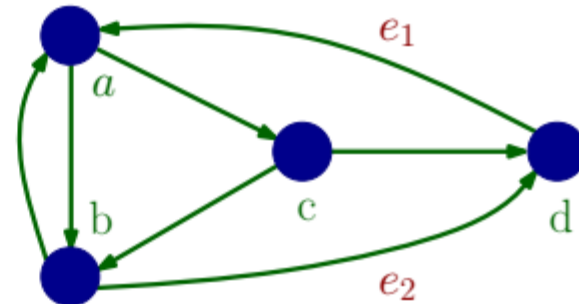
Directed Graphs (digraphs)

$$G = (V, E)$$

V is set of vertices

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(ordered pairs of V)

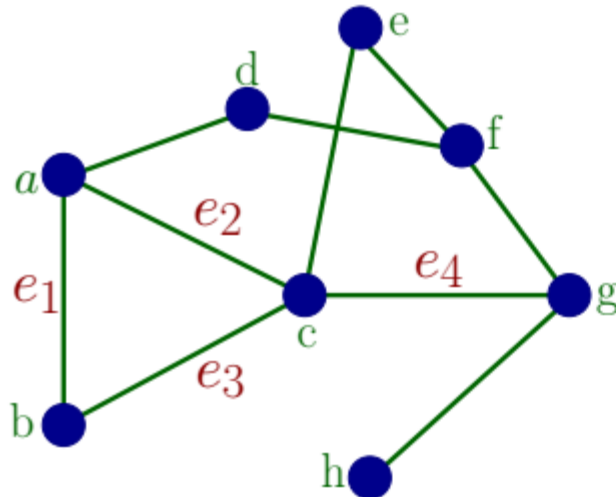


Undirected Graph Representation: Adjacency Matrix

Represent undirected $G = (V, E)$ with an **adjacency matrix** A_G

- Fix an arbitrary ordering of V
- One row for each vertex in V
- One column for each vertex in V

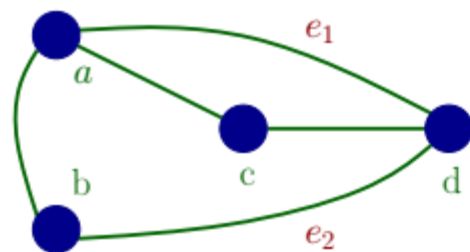
$$A_{ij} = \begin{cases} 1 & \text{if } (v_i, v_j) \in E \\ 0 & \text{if } (v_i, v_j) \notin E \end{cases}$$



$A_G =$

	a	b	c	d	e	f	g	h
a	0	1	1	1	0	0	0	0
b	1	0	1	0	0	0	0	0
c	1	1	0	0	1	0	1	0
d	1	0	0	0	0	1	0	0
e	0	0	1	0	0	1	0	0
f	0	0	0	1	1	0	1	0
g	0	0	1	0	0	1	0	1
h	0	0	0	0	0	0	1	0

Undirected Graph Representation: Adjacency Matrix



$$A = \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}$$

Let A be the adjacency matrix of a simple graph graph on n vertices.

ICP 14-22

How many entries are there in A ?

ICP 14-23

How many 1's are there in A ?

ICP 14-24

How many 0's are there in A ?

ICP 14-25

What are diagonal entries of A ?

ICP 14-26

How many 1's are there in row corresponding to vertex v ?

ICP 14-27

How many 1's are there in column corresponding to v ?

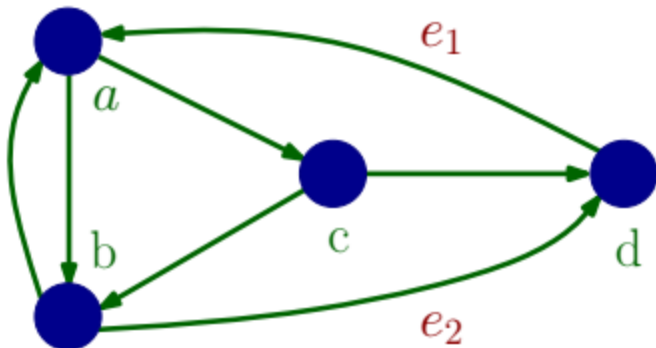
Directed Graph Representation: Adjacency Matrix

Digraph $G = (V, E)$ is a relation on V

Represent G with an **adjacency matrix** A_G

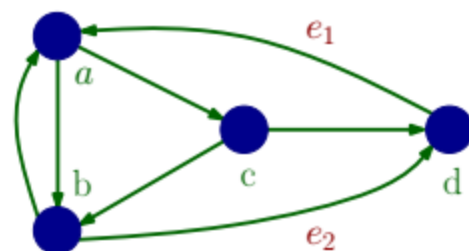
- Fix an arbitrary ordering of V
- One row for each vertex in V
- One column for each vertex in V

$$A_{ij} = \begin{cases} 1 & \text{if } (v_i, v_j) \in E \\ 0 & \text{if } (v_i, v_j) \notin E \end{cases}$$



$$A = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix}$$

Directed Graph Representation: Adjacency Matrix



$$A = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix}$$

Let A be the adjacency matrix of a simple digraph graph on n vertices.

ICP 14-28 How many entries are there in A ?

ICP 14-29 How many 1's are there in A ?

ICP 14-30 How many 0's are there in A ?

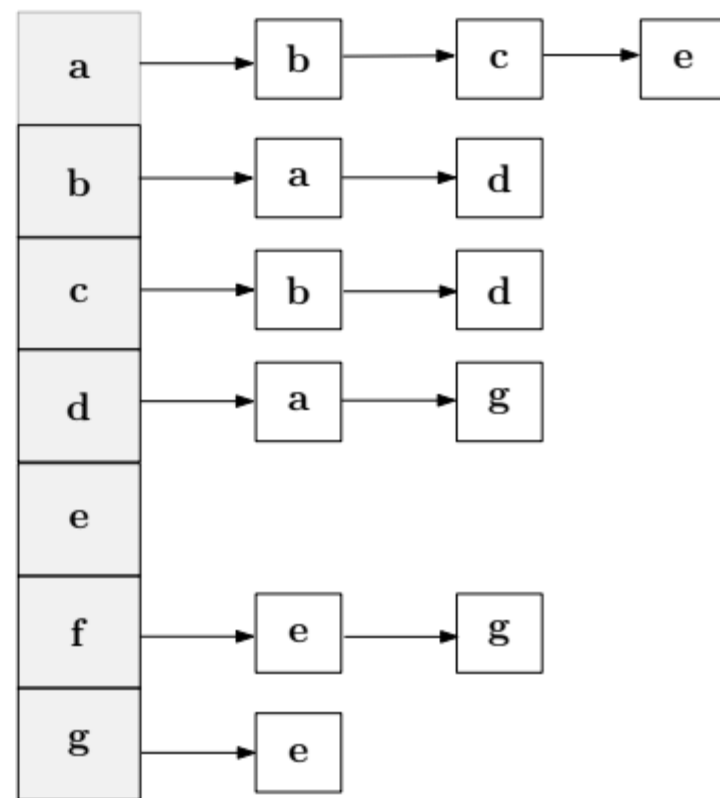
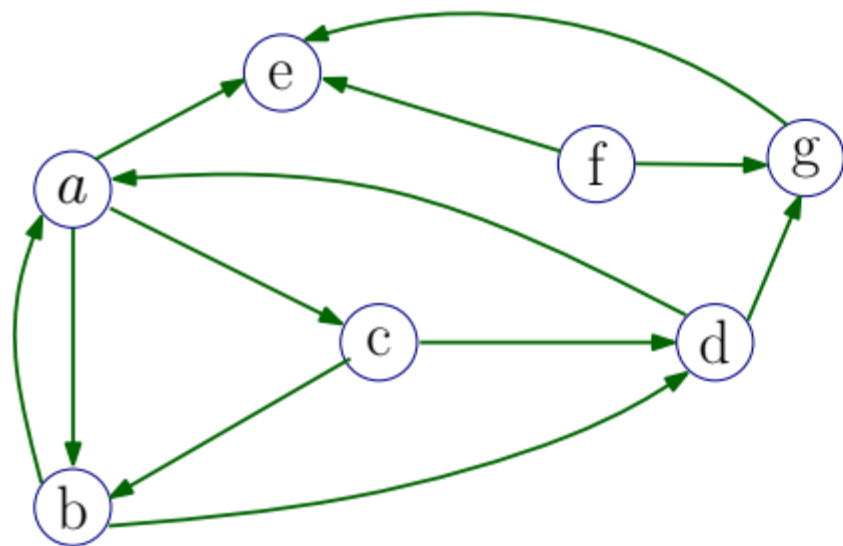
ICP 14-31 What are diagonal entries of A ?

ICP 14-32 How many 1's are there in row corresponding to vertex v ?

ICP 14-33 How many 1's are there in column corresponding to v ?

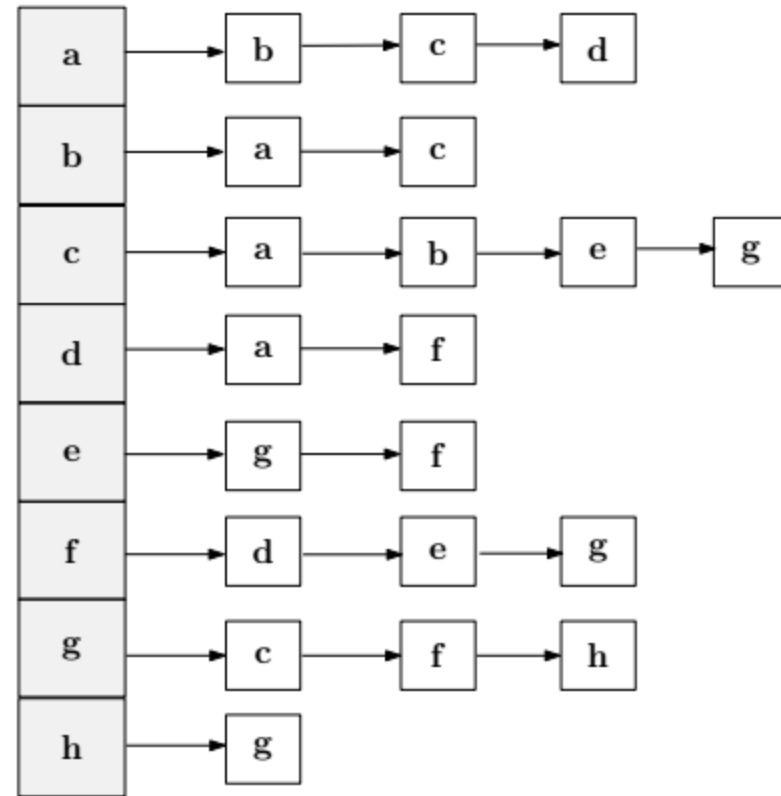
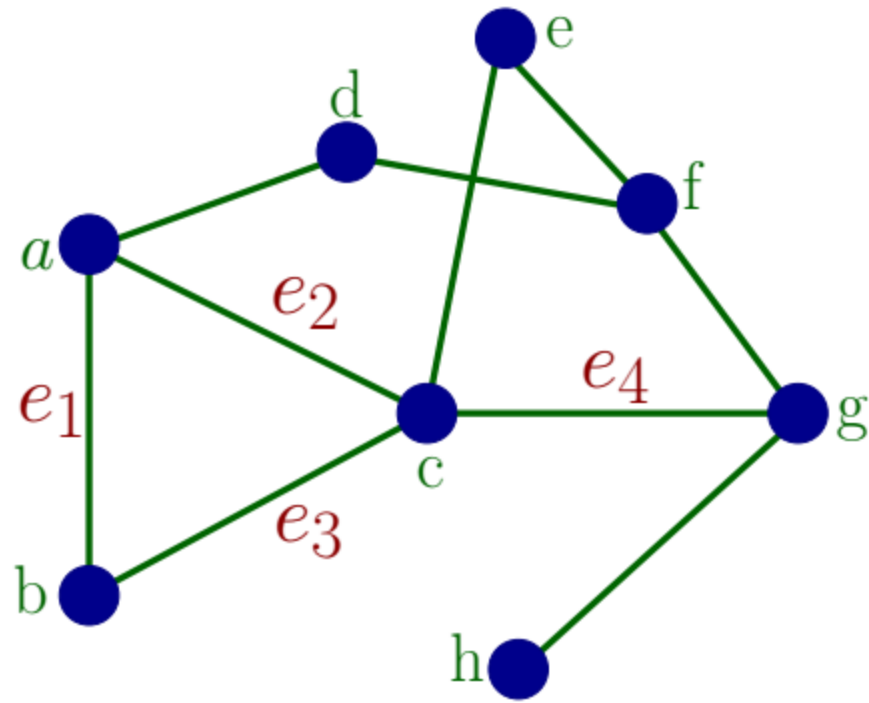
Graph Representation: Adjacency List

Represent digraph by listing neighbors of each vertex

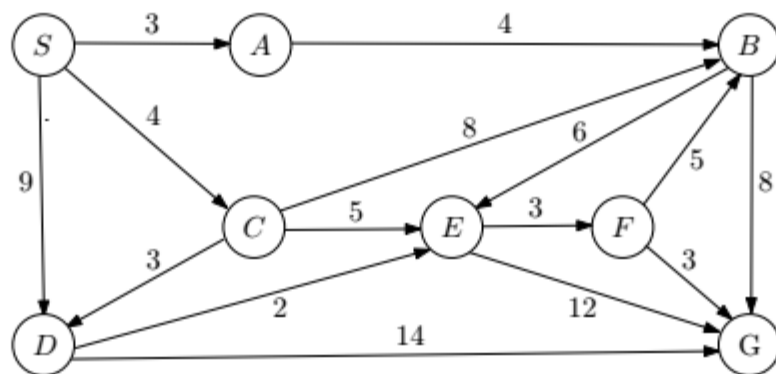


Graph Representation: Adjacency List

Represent undirected graph by listing neighbors of each vertex



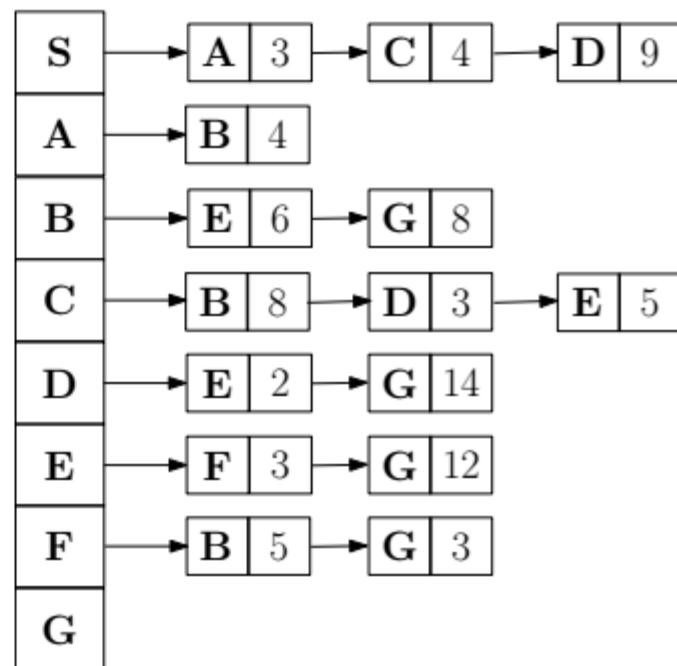
Weighted Graph Representation



Weighted Adjacency Matrix

	S	A	B	C	D	E	F	G
S	0	3	0	4	9	0	0	0
A	0	0	4	0	0	0	0	0
B	0	0	0	0	0	6	0	8
C	⋮							
D								
E								
F								
G								

Weighted Adjacency Lists



Subgraph

$H = \underline{(V', E')}$ is a **subgraph** of $G = \underline{(V, E)}$, if

$$V' \subseteq V \quad \text{AND} \quad E' \subseteq E$$

Denoted as $H \subseteq G$

